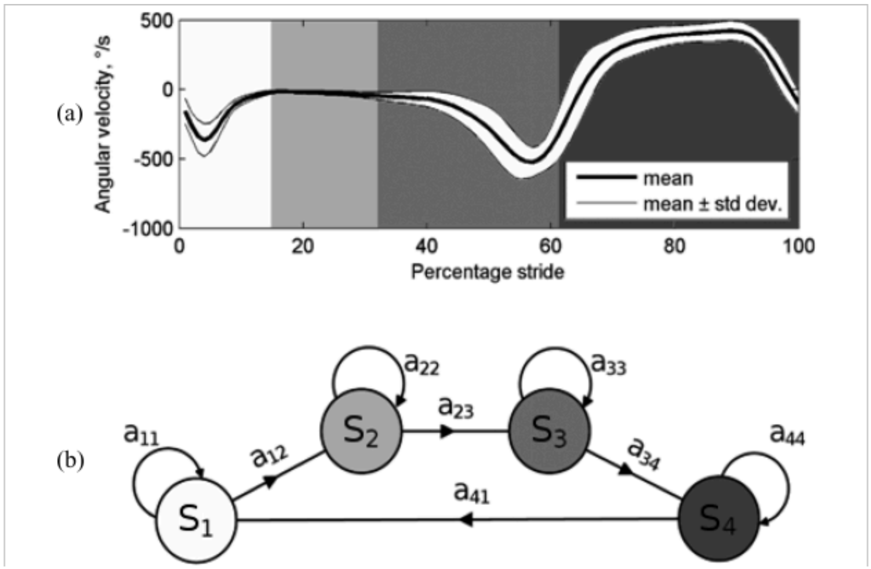
Gait analysis methods:

- Fractal analysis (fractal dim, Hurst exp)
- Recurrence quantification analysis
    1. Fractal dim: R/S, roughness-length, variogram, and wavelets analysis
    2. RQA: delayed coordinate embedding; embedding dimension, time delay, and false nearest neighbor were determined(recurrence, determinism, ratio, entropy, max line, trend, and spatio-temporal entropy)
- Lyapunov exponent
  Short-term LE is a better measure than long-term
- Entropy measures(on Stride interval time series)
    1. Unlike ApEn and SampEn, SVDEn does not require the specification of tolerance and embedding dimension parameters, relying only on the scale parameter.
    2. DistEn demonstrated high parametric consistency and low sensitivity to parameter changes compared to ApEn and SampEn.
- Nonlinear dynamics (phase space reconstruction, Poincare maps)
    1. ankle position trajectory as phase space: dimension number was set according to the false nearest neighbors' method, and the time delay was set based on the detection of the first local minimum of the mutual information
       (Before carrying out the t-tests, its underlying assumptions of normality and homogeneity of variance were tested using the Shapiro–Wilk's test and Leven's test, respectively.)
- Statistical/mathematical analysis
    1. Hjorth parameters, energy, SMA (Contrast, homogeneity, correlation, uniformity, and maximum probability of the co-occurrence matrix are used as features of the user gait)
       NARX (Nonlinear AutoRegressive eXogenous) Model (Predict the next value using the NARX model, Compute FIT over a sliding window, Classify walk as normal or abnormal based on FIT threshold)
- Hidden Markov models- insole sensors
    1. FFT->PCA->HMM (Baum-Welch for parameter optimization, model similarity through log-likelihood and identification rates)
        - Each class learned separately for each subject
    2. GRF(at 4 different locations)- gait phases, transition probabilities studied, conditional posterior modeling
    3. HS (stride time, cadence, and normalized stance); FF & HO (stride length and walking speed using strap-down navigation techniques)

(ref: Wavelet transformation and peak detection algorithms were applied to the angular velocity; estimation of gait spatial parameters was then performed using simple biomechanical models of the lower limb)



- addressing the limitations of the standard Viterbi algorithm with the Segmental Viterbi (STV) algorithm
- improves the efficiency of online decoding by dynamically adjusting to observed feature vectors and optimizing state transitions

- ~~Linear Discriminant Analysis (Classification in high dimension)~~
- Cross-Recurrence Quantification Analysis (CRQA)- coordination between left & right foot
- Empirical Mode Decomposition (EMD)

(decompose gait rhythm time series into intrinsic mode functions (IMFs), Kendall's coefficient of concordance (W), and the ratio of energy change (RE) were calculated for different IMFs)

Tests:

- UPDRS(Unified Parkinson's Disease Rating Scale), and SPPB(The Short Physical Performance Battery), to validate/compare our results
- Future fall-risk prediction systems estimate the user's fall risk in an offline test such as Timed Up and Go (TUG), Berg Balance Scale (BBS), Sit To Stand (STS), and One-Leg Stand (OLS)

Tools:(Python libraries for fractal methods on physiological data)

- NeuroKit(ECG,PPG,EMG,EEG)
- AntroPy
- MNEPython(EEG)

Papers

https://www.frontiersin.org/articles/10.3389/fspor.2022.893745/full
We used time-delay embedding to reconstruct the position trajectories to a phase space that represents the states of the ankle dynamics. Based on the phase space trajectory, a recurrence plot was constructed and two RQA variables, the percent determinism (%DET) and the percent laminarity (%LAM), were derived from the recurrence plot to quantify the ankle dynamics.
In the frontal plane, the %LAM in the CAI group was significantly lower than that in the control group ($p < 0.05$. effect size = 0.86). This indicated that the ankle dynamics in individuals with CAI are less likely to remain in the same state. No significant results were found in the %DET or the sagittal plane.
gait variability
Ref: Linear methods, such as standard deviation cannot detect the variation in temporal sequence.
nonlinear methods such as sample entropy
(compare the complexity of the center of pressure trajectories during a single-leg balance task between those with and without CAI)

https://www.sciencedirect.com/science/article/pii/S1746809423008728?via=ihub
ApEn, SampEn, SVDEn
variability of stride interval time
average maximum finite-time Lyapunov exponents to quantify the local dynamic stability of human walking kinematics
Holder exponent approach - find that the fractal nature of the stride interval fluctuations is more pronounced under deviations from the average gait
Spectral analysis showed that the ratio of the stride time fluctuations on relatively large time scales to the fluctuations on shorter time scales decreased with age
Pearson's correlation coefficient was computed, and the values ranged from 0.45 to 0.69. However, a two-tailed *t*-test indicated that the correlations are not significant ($p < 0.05$). One concludes that the SVDEn does not provide the same information as the CV index.
(ref)distribution entropy

https://link.springer.com/article/10.1007/s10286-002-0044-8
HR & PTT(Pulse Transit Time):

The fractal dimension of the time curve (FD) was calculated with the aid of four different methods: R/S, roughness-length, variogram, and wavelets analysis.
RQA: The embedding dimension, time delay, and false nearest neighbor were determined. On this basis, the RQA variables were computed: recurrence, determinism, ratio, entropy, maxline, trend, and spatiotemporal entropy.

https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=7422732: empirical mode decomposition
https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8051436/
Entropy analysis - review paper

Math:https://onlinelibrary.wiley.com/doi/10.1155/2018/4750104 - simulation of abnormal walk datasets present(co-occurrence matrix features used)

https://scholar.google.co.in/scholar_url?url=https://www.academia.edu/download/39601828/Medical_Engineering___Physics20151101-8477-1szphmn.pdf&hl=en&sa=X&ei=RJ19Zt70OJWx6rQPnaGcgAg&scisig=AFWwaeZneDhIeDKrSAGmn5CaJLIW&oi=scholarr
Lyapunov exponents

Existing Methods

- **Approximate Entropy (ApEn):** Measures the conditional probability that similar patterns remain similar at the next point, requiring parameters like pattern length (m) and tolerance (r).
- **Sample Entropy (SampEn):** An improvement over ApEn, excluding self-matches in probability calculations, but still dependent on the selection of parameters.
- **Singular Value Decomposition (SVD) Entropy**
- **Concept:** SVD is a method to quantify the distribution of information induced by a linear transformation, using an analog of Shannon entropy to measure complexity.
- **SVD Process:**
  - **Matrix Construction:** Create a matrix M_X(t_i; n) from lagged sequences of the time series.
  - **SVD Factorization:** Decompose the matrix into U, Λ, and V matrices, where Λ contains the singular values.
  - **Entropy Calculation:** Normalize the singular values and compute the entropy based on their distribution.
  - **SVD Entropy Calculation**

1. **Normalize Singular Values:**

$$\lambda_j^*(t_i;n) = \frac{\lambda_j(t_i;n)}{\sum_{j=1}^{n} \lambda_j(t_i;n)}$$

2. **Entropy Formula:**

$$S(t_i;n) = -\sum_{j=1}^{n} \lambda_j^*(t_i;n) \ln(\lambda_j^*(t_i;n))$$

3. **Normalized Entropy:**

$$SVDEn(t_i;n) = \frac{-1}{\ln(n)} \sum_{j=1}^{n} \lambda_j^*(t_i;n) \ln(\lambda_j^*(t_i;n))$$

   - Ensures values between 0 and 1, where 1 indicates no correlations, and values less than 1 indicate the presence of correlations.

Hurst exponent

Divide the time series into segments of equal length.

Calculate the range (R) and standard deviation (S) for each segment.

Compute the rescaled range R/S for each segment.

Plot log(R/S)versus log(n), where n is the segment length.

The slope of the line fitted to these points is the Hurst exponent H.

- H<0.5: Anti-persistent behavior (highly volatile, changes direction frequently).
- H=0.5: Random walk (no correlation, typical of white noise).
- H>0.5: Persistent behavior (trends, less volatile, positive correlation).

Fractal dimension

Cover the time series plot with a grid of boxes of size ϵ

Count the number of boxes $N(\epsilon)$ that contain part of the time series.

Repeat for different values of ϵ.

Plot $\log(N(\epsilon))$ versus $\log(1/\epsilon)$

The slope of the line fitted to these points is the fractal dimension D.

- A higher fractal dimension indicates a more complex and irregular pattern.

Lyapunov exponent:

- Identify Neighboring Trajectories: In the reconstructed phase space, identify pairs of neighboring trajectories. Calculate the distance between these points.
- Track Divergence: Track the divergence of these trajectories over time. The Lyapunov exponent is the average exponential rate of divergence of initially close trajectories.

$\lambda = \lim_{t\to\infty} \frac{1}{t} \ln \frac{d(t)}{d(0)}$

Where $d(t)$ is the distance between two trajectories at time t and $d(0)$ is the initial distance.

- Positive Lyapunov Exponent: Indicates chaotic behavior and sensitivity to initial conditions, suggesting instability or unpredictability in the gait pattern.
- Near-Zero Lyapunov Exponent: Indicates regular, periodic, or stable behavior.
- Negative Lyapunov Exponent: Rare in practical gait analysis, but would suggest converging trajectories.

Threshold Setting: Establish thresholds for the Lyapunov exponent based on normal and abnormal gait patterns.

DisEn:

Template Matching:

- Choose an embedding dimension m and a tolerance r.
- Form vectors Xi=[x(i),x(i+1),...,x(i+m−1)] from the time series x.
- Calculate the distances between these vectors using an appropriate metric (e.g., Chebyshev distance).

Probability Distribution:

- Calculate the number of template matches that fall within the tolerance r.
- Normalize these counts to obtain a probability distribution.

Entropy Calculation:

- Compute the Shannon entropy of the obtained probability distribution.

- DistEn is defined as: DistEn=-∑pilog(pi) where pi is the probability of the i-th template match.

Sample Entropy (SampEn) Calculation:

- For each coarse-grained time series, calculate the sample entropy.
- Sample entropy (SampEn) is defined as:

$$\text{SampEn}(m, r, N) = -\ln\left(\frac{A}{B}\right)$$

where A is the number of template matches of length $m+1$ and B is the number of template matches of length m.

Plot the SampEn values against the different scales to create the MSE curve.

RQA: recurrence rate, laminarity, determinism, entropy